

Implementation Notes: Speech Understanding Pipeline

SPEECH UNDERSTANDING ASSIGNMENT-2

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Part I: Robust Code-Switched Transcription (STT)

Non-Obvious Design Choice: Dynamic Spectral Subtraction Floor

When implementing the noise reduction for the classroom audio in Task 1.3, I opted to use a dynamic spectral subtraction floor rather than a hard cutoff. Standard spectral subtraction often introduces "musical noise" (artificial ringing) when aggressively filtering reverberant classroom environments. To mitigate this, I implemented a smoothing factor that dynamically adjusts the over-subtraction penalty based on the localized Signal-to-Noise Ratio (SNR) of each frame. This preserves the high-frequency fricatives in the Hindi speech (which are often destroyed by aggressive noise reduction) while maintaining an F1 > 0.85 for the LID model.

Part II: Phonetic Mapping & Translation

Non-Obvious Design Choice: Hybrid Rule-Based & Dictionary G2P Fallback

Converting Hinglish code-switched text directly to the International Phonetic Alphabet (IPA) in Task 2.1 is notoriously difficult because standard G2P models are exclusively trained on English phonology. To solve this, I designed a hybrid approach: I utilized a customized dictionary for technical terms (e.g., "stochastic", "cepstrum") to ensure their English phonetic roots were preserved, but dynamically fell back to a custom rule-based transliteration engine for the Hindi connective tissue. This ensured that code-switched boundaries weren't hallucinated as mispronounced English words, greatly improving the fidelity of the Gondi semantic translation.

Part III: Zero-Shot Cross-Lingual Voice Cloning (TTS)

Non-Obvious Design Choice: Dimensionality Adaptation for Windows Compatibility

The most significant architectural hurdle occurred during the speaker embedding extraction in Task 3.1. The official SpeechT5 voice cloner strictly expects a 512-dimensional x-vector. The standard model for this (`speechbrain/spkrec-xvect-voxceleb`) relies on the `k2` library, which requires complex C++ compilation and frequently fails on Windows. To ensure the pipeline was fully robust and executable on Windows environments, I made the non-obvious choice to fall back to the 192-dimensional ECAPA-TDNN model (`spkrec-ecapa-`

voxceleb`). To bridge the architectural gap, I implemented a custom adaptation layer that manually zero-pads the 192-dim latent vector to 512 dimensions. While this slightly degrades the clone's specificity, it bypasses the Windows `k2` fatal crash entirely, ensuring the system can reliably generate the 10-minute Gondi audio.

Part IV: Adversarial Robustness & Spoofing Detection

Non-Obvious Design Choice: Max-Feature-Map (MFM) Activation for the LightCNN

For the Anti-Spoofing Countermeasure (Task 4.1), distinguishing between bona fide human audio and high-quality SpeechT5 synthesis is incredibly challenging. Instead of using standard ReLU activations in the LightCNN, I made the design choice to implement Max-Feature-Map (MFM) activations based on Network-in-Network principles. MFM inherently acts as a feature selector by suppressing low-activation neurons and promoting competitive learning between filter channels. This allowed the model to focus specifically on the high-frequency spectral artifacts generated by the TTS vocoder (the primary indicators of spoofed audio) rather than being distracted by the underlying phonetic content, drastically stabilizing the Equal Error Rate (EER) evaluation.